R0, Fc, M1 and C0 — source-faithful publication units

Scope: BR3-008 (R0), BR3-023 (Fc), BR3-044 (M1), BR3-027 (C0) and the shared intervention note, with the existing BR3-012 units unchanged. Release candidate for review; not deployed. Each English rendering is shown with its Japanese original and its mandatory note.

> **Scope: Publication layer (source-faithful)**
> Shows the Japanese original, the approved English label and gloss, and the mandatory note. Necessary-condition edges of the English reconstruction (E2-001, E2-002) are not drawn, and are neither asserted nor denied.

Publication units

R0 相互作用 Interaction

物理的結合があり、状態・過程に影響する。

There is a physical coupling, and it affects a state or process.

> Causal, not mere correlation. Intervention is the map's test, not part of the definition. See note.
>
> Note. 影響する is kept as 'affects'. It is a causal relation: the Japanese map warns against accepting a causal coupling from correlation alone. The map's test asks whether intervening on the coupling changes the partner's state or process. That test is not written into the definition, so intervention-sensitivity is not presented as a condition every R0 instance must meet. The English reconstruction uses the label 'causal coupling' and retains intervention-sensitivity as a conditional reconstruction-layer proposal. The label and the proposal are not part of the Japanese R0 definition. 状態・過程 is rendered 'a state or process'; the source does not say whether one or both are meant. The link from R0 to R1 is a comparison in the Japanese map; it neither establishes nor denies that R1 requires R0.
>
> Intervention. In the source-faithful publication layer, intervention is treated as an assessment route, not automatically as a constitutive condition of R0, Fc or M1. It can test a relation under a declared intervention design and comparison conditions. A relation is not generally absent merely because no intervention was performed or because its causal path was not fully traced. A null result from a single intervention does not, by itself, establish the general absence of the relation; its evidential significance depends on the intervention design and declared comparison conditions. Being open to intervention is distinct from having been established by a particular intervention. The English reconstruction may retain layer-specific intervention requirements, including D-H4 for Fc.

Constitutive conditions (publication)
Physical coupling; causal influence on a state or process

▶ **Audit conditions (publication scope)**

Fc 因果役割的機能 **Causal-role function**

指定した能力・過程への寄与。

Contribution to a specified capacity or process.

> More than correlation; no uniform single-intervention, full-path, indispensability or component requirement. Intervention is an important assessment route. See note.
>
> Note. 寄与 is read here as substantive causal involvement in the specified capacity or process: more than correlation, co-occurrence or loose association. It does not require that a single intervention shows a difference, that the whole causal path is traced, that the contributor is indispensable, or that the contributor must be a component of Q. Intervening on the bearer is an important way of assessing a contribution. The depth of a contribution claim is distinguished from the strength of evidence supporting it. Whether 寄与 also suggests a constitutive or foundational role is left as an audit question, not part of the definition. Current contribution is distinct from organizational closure (Fo) and selection history (Fs). The English reconstruction keeps an intervention requirement (decision D-H4) as its own theoretical choice; it is not a condition of the Japanese Fc.
>
> Intervention. In the source-faithful publication layer, intervention is treated as an assessment route, not automatically as a constitutive condition of R0, Fc or M1. It can test a relation under a declared intervention design and comparison conditions. A relation is not generally absent merely because no intervention was performed or because its causal path was not fully traced. A null result from a single intervention does not, by itself, establish the general absence of the relation; its evidential significance depends on the intervention design and declared comparison conditions. Being open to intervention is distinct from having been established by a particular intervention. The English reconstruction may retain layer-specific intervention requirements, including D-H4 for Fc.

Constitutive conditions (publication)

Specified capacity or process Q; substantive causal contribution to Q

▶ **Audit conditions (publication scope)**

M1 関連性 **Q-relative difference-making**

差が指定Qへの帰結を変える。$\mathrm{Rel}(d;Q|\Sigma,\mathscr{D},M,\tau)$。

A difference changes the consequences for a specified Q. $\mathrm{Rel}(d;Q|\Sigma,\mathscr{D},M,\tau)$

Relative to a specified Q and to Σ, $\mathscr{D}$, M, τ (M = mechanism). Not benefit. Intervention on d is an assessment route, not part of the Japanese definition. See note.

Note. The relation holds relative to a specified Q and to the declared boundary Σ, comparison set $\mathscr{D}$, mechanism M and time scale τ; a change in any of these is a different claim. The M in the formula is the mechanism, not the M group of the map. 関連性 here is not usefulness or benefit and does not by itself imply meaning or representation. Being comparable within $\mathscr{D}$ is not the same as an intervention being definable. Intervening on the difference d is different from intervening on Q directly; the map's test asks whether intervening on d changes the consequence for Q. What is intervened on, the path of action and what is evaluated are audited separately. Observation, comparison and intervention are different routes of assessment, and they are not treated as giving equal evidence. Neither a complete intervention nor full identification of the causal path is required for the relation to hold. The English reconstruction's typed Q and intervention_basis are English-side method settings, and its placement of this relation outside the M group does not change where the Japanese map places M1.

Intervention. In the source-faithful publication layer, intervention is treated as an assessment route, not automatically as a constitutive condition of R0, Fc or M1. It can test a relation under a declared intervention design and comparison conditions. A relation is not generally absent merely because no intervention was performed or because its causal path was not fully traced. A null result from a single intervention does not, by itself, establish the general absence of the relation; its evidential significance depends on the intervention design and declared comparison conditions. Being open to intervention is distinct from having been established by a particular intervention. The English reconstruction may retain layer-specific intervention requirements, including D-H4 for Fc.

Constitutive conditions (publication)
A difference d; a specified Q; d changes the consequences for Q; relative to declared Σ (boundary), $\mathscr{D}$ (comparison set), M (mechanism) and τ (time scale)

▶ **Audit conditions (publication scope)**

C0 受動的安定化 Passive stabilization

物理化学的な応答によって変化が減衰する。

Change decays through physicochemical responses. (‘passive’ ≠ ‘no pathways’; see note)

> 'Passive' ≠ 'no pathways' or 'pathways unidentified'. Not exclusive with C1/C2. See note.
>
> Note. 'Passive' does not require that the system has no sensing or action pathways, that such pathways cannot be told apart, or that they have not been identified; whether they exist, can be distinguished and have been identified are separate questions. C0 is not exclusive with C1 (negative feedback) or C2 (specific regulation): they can be attributed in the same system, for example where binding relaxations occur alongside feedback-based adaptation. A stable state should not be read back as a set point or goal, passive relaxation should not be equated with sensing, and whole-system stability is not inferred. What mechanistic restriction 'passive' places on an individual decay route is left open.

Constitutive conditions (publication)
A change; decay of the change; through physicochemical responses

▶ **Audit conditions (publication scope)**

BR3-012 units (existing publication, unchanged)

R4 コード／体系的対応

R3的規則が多数の入力に体系的・組合せ的に適用される。

An R3-like rule is applied to many inputs systematically and combinatorially.

Source: 体系的・組合せ的. Both required, or simply listed? Left open in the source; "and" here does not decide it. Author's provisional translation; comments welcome.

Note on the translation. The Japanese original reads 「体系的・組合せ的」. The source leaves open whether every case must be both systematic and combinatorial, or whether the phrase simply lists the two; the English "and" is not meant to settle that question. This rendering is the author's current choice, made after comparing alternatives, and has not yet been reviewed by native English speakers. Other readings and translations are welcome, especially from native speakers and researchers in related fields. Any revision will keep the original text and the decision history.

R4-SYS (EN-ICM-0059)

R3的規則が多数の入力に体系的・組合せ的に適用される。 (source of BR3-012)

An R3-like correspondence rule is applied systematically across multiple inputs. Whether the rule satisfies every R3-BASE condition is not asserted.

Systematic axis of R4 (体系的); relation to the combinatorial aspect (組合せ的) is undecided in the source.

Axis note. This English-side axis carries the systematic aspect (体系的) of the Japanese R4. The Japanese R4 also names a combinatorial aspect (組合せ的; see R4-COMB). The source does not settle whether both are required together; showing them as two axes is not a ruling on that question.

R4-COMB (EN-ICM-0057)

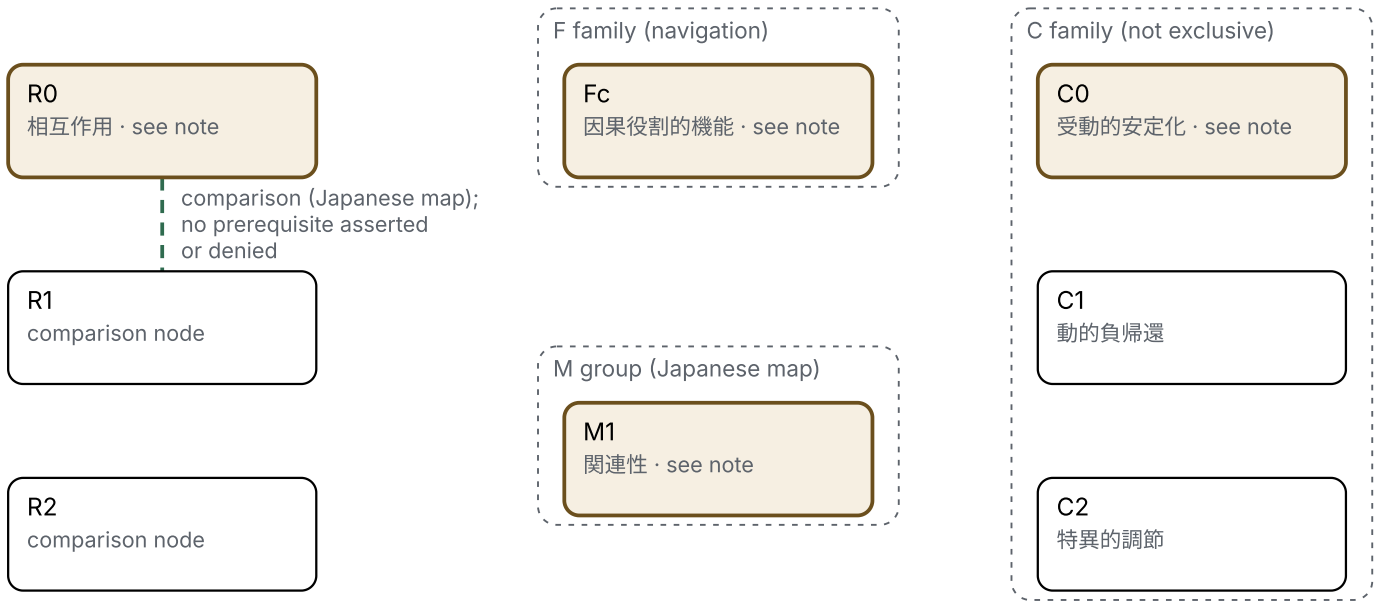
R3的規則が多数の入力に体系的・組合せ的に適用される。 (source of BR3-013)

A correspondence is organized over recombinable elements or structures such that combinations participate in the mapping.

Combinatorial axis of R4 (組合せ的); without the R3-like rule (target-side); relation to 体系的 undecided.

Axis note. This English-side axis carries the combinatorial aspect (組合せ的) of the Japanese R4. As a target-side scope choice it does not carry the R3-like rule (BR3-013). The source does not settle whether the combinatorial and systematic aspects are required together.

Publication graph



Legend

R0 相互作用 Interaction

> Causal, not mere correlation. Intervention is the map's test, not part of the definition. See note.

Fc 因果役割的機能 Causal-role function

> More than correlation; no uniform single-intervention, full-path, indispensability or component requirement. Intervention is an important assessment route. See note.

M1 関連性 Q-relative difference-making

> Relative to a specified Q and to Σ, $\mathscr{D}$, M, τ (M = mechanism). Not benefit. Intervention on d is an assessment route, not part of the Japanese definition. See note.

C0 受動的安定化 Passive stabilization

> 'Passive' ≠ 'no pathways' or 'pathways unidentified'. Not exclusive with C1/C2. See note.

Dashed green: Source comparison; prerequisite neither asserted nor denied.

Appendix — English-reconstruction items not shown as Japanese source conditions

E2-001: R1 requires R0 (English reconstruction layer). Edge note. E2-001 (R1 requires-condition R0, entailment necessary) is part of the English reconstruction R condition order. Its source support is not adjudicated (LDG-0011@v0.2). In the publication layer it is neither shown as a necessary condition nor denied.
Evidence: NOT-ASSESSED (source support unadjudicated; tracked in LDG-0011@v0.2) · display: HIDDEN-LISTED

E2-002: R2 requires R1 (English reconstruction layer). Edge note. E2-002 (R2 requires-condition R1, necessary) carries the R condition order further; through E2-001 it carries R0 conditions to R2 and beyond (E2-003, E2-008, E2-010, and E2-004 in reconstruction). Tracked in LDG-0011@v0.2.
Evidence: NOT-ASSESSED (tracked in LDG-0011@v0.2) · display: HIDDEN-LISTED

QREL (EN-ICM-0053) address: QREL is placed outside the M family in the English reconstruction (H11-S1). Architecture note. The English reconstruction places Q-relative difference-making outside the M family. This does not change where the Japanese map places M1 (group M).
Evidence: TARGET-SIDE-PROPOSAL (H11-S1) · display: HIDDEN-LISTED

Data: frozen-inputs/english_publication_overlay_v0.7.tsv (CSV reader) · frozen-inputs/four_bridge_condition_provenance_v0.5.tsv · research-audit/english_data_draft_v0.8.tsv. Render check only; canonical effect NONE.